

Freight Rate Prediction Challenge

Machine Learning Assessment Report

Final solution: CatBoost Regression with Log1p target transformation

Submission repository: github.com/sharmeenaehsaan/freight-rate-prediction

1. Objective

The objective was to build a regression model that predicts the posted freight rate for transportation loads. The development dataset contains 48,000 labeled loads, while the assessment validation dataset contains 12,000 loads without the target. A second fixed dataset contains 31 December 2025 loads for the required prediction chart.

2. Data and preprocessing

The main predictive inputs include pickup and delivery locations, pickup and delivery coordinates, distance, equipment type, weight, date, market index, and quote signal. Duplicate rows and duplicate load IDs were checked during exploratory analysis.

- Negative weight values were treated as sign errors and converted with absolute value.
- Missing numerical values were imputed with medians calculated from the training portion used for model development.
- The date was expanded into year, month, day, day of week, and day of year.
- A route feature was created as pickup -> delivery so the model could learn lane-specific patterns.

3. Model development

Several candidate approaches were tested, including a baseline CatBoost model, geographic-distance variants, direct-target regression, Log1p target modeling, weighted training for high-rate observations, route-history features, high-rate analysis, and multi-stage approaches. The Log1p CatBoost approach provided the strongest internal development result among the retained experiments.

4. Selected model

Parameter	Value
Algorithm	CatBoostRegressor
Target	log1p(posted_rate)
Iterations	282
Learning rate	0.05
Depth	8
Loss function	RMSE
Random seed	42

Categorical features	pickup, delivery, equipment, route
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The model predicts the transformed target and converts predictions back to the original rate scale with expm1. The final model was retrained on all 48,000 development rows before generating the 12,000 assessment predictions.

5. Development split and results

Model selection used an 80/20 random development split. The split contained 38,400 training rows and 9,600 validation rows. The split was stratified with respect to extreme high-rate observations (rates at or above \$7,500).

Metric	Development validation
MAE	\$93.15
RMSE	\$588.71
R2	0.8461

These metrics are internal development metrics. They are not the official scores for the 12,000 assessment validation loads because the assessment validation file does not contain the actual posted_rate target.

6. Final validation submission

The final model generates validation_predictions.csv with exactly two columns: load_id and predicted_rate. The file contains predictions for all 12,000 validation loads. The provided scorer accepted the submission file with the message: “Validated 12,000 final predictions.”

7. December 2025 fixed-route predictions

The fixed December inputs represent the Lexington -> Fort Wayne route, distance 360, Dry Van equipment, and weight 32,000 for December 1 through December 31, 2025. The final model generated all 31 predicted rates, and the scorer confirmed: “Validated 31 fixed December predictions.”

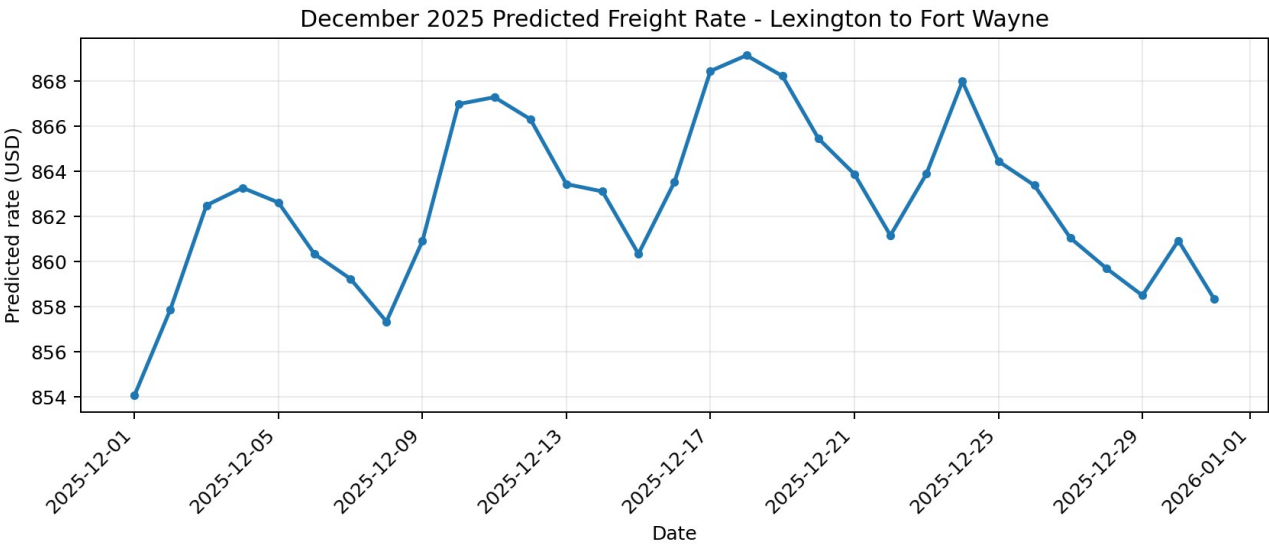


Figure 1. Final December prediction curve generated from the completed assessment input file.

8. December prediction summary

Statistic	Predicted rate
Minimum	\$854.07
Maximum	\$869.15
Mean	\$862.71
Median	\$863.12

9. Reproducibility and scoring

The final workflow is contained in `train_final_model.py` and `generate_december_predictions.py`. The assessment scorer was executed with:

```
python score.py --predictions validation_predictions.csv --december-predictions data/december-chart-inputs.csv
```

The scorer created `scorer_results/candidate_december.png`. Official metrics for the hidden-target validation set are produced by the assessment platform after submission.

10. Deliverables

- GitHub repository containing source code, datasets, experiment records, model, predictions, scorer output, README, and requirements.
- `validation_predictions.csv` containing the 12,000 final predictions.
- Completed December input file with `predicted_rate` values.
- `candidate_december.png` generated by the provided scorer.
- This report describing the data preparation, model development, validation split, and December prediction chart.

End of report